

Income tax to include in the net present value calculation

Renewable energy investment decision: additional power purchase agreements compared with direct investment in generation capacity. Corporate income tax rate 27 per cent. Effective rate on capital gains for a company 21.6 per cent, being the 80 per cent inclusion rate applied at 27 per cent. **Apply this only where the cash flows are stated after tax and the discount rate is a post-tax weighted average cost of capital. Where pre-tax cash flows are discounted at a pre-tax rate, none of the adjustments below are made.**

Cash flow line	Additional power purchase agreements	Direct investment in generation capacity	Treatment in the net present value calculation
Electricity benefit, being the cost of grid electricity avoided	Included in taxable income	Included in taxable income	Include at the amount after tax, that is the amount less 27 per cent
Payments for electricity under the agreement	Deductible in full under section 11(a)	Not applicable	Include at the amount after tax, that is the amount less 27 per cent
Wheeling, network and administration charges	Deductible in full under section 11(a)	Deductible in full under section 11(a) where generation is off site	Include at the amount after tax, that is the amount less 27 per cent
Operating, maintenance and insurance costs	Not applicable	Deductible in full under section 11(a)	Include at the amount after tax, that is the amount less 27 per cent
Proceeds from surplus electricity sold, wheeled or credited	Included in taxable income when it accrues	Included in taxable income when it accrues	Include at the amount after tax, that is the amount less 27 per cent
Initial capital expenditure	Not applicable	Not deductible	Include the full amount as a cash outflow. No tax adjustment is made to this line
Capital allowances	None. No asset is owned, and sections 12B, 12BA and 11(e) each require ownership	Section 12B at 50 per cent, 30 per cent and 20 per cent over three years, or 100 per cent in year one for photovoltaic solar not exceeding 1 megawatt. Section 12U at 100 per cent on new roads and fences. Section 13quin at 5 per cent on new buildings. Section 11(e) on the grid connection and transformers	Not a cash flow in themselves. Include a separate tax saving equal to the allowance multiplied by 27 per cent
Land	Not applicable	No allowance is available under any section	Include the full amount as a cash outflow. Relief arises only as base cost on a later disposal
Amounts paid in advance under the contract	Section 23H spreads the deduction over the period to which the expenditure relates	Section 23H spreads the deduction over the period to which the expenditure relates	Include the full amount as a cash outflow when paid, and spread the deduction. A difference in timing only
Interest on funding	Not applicable	Deductible under section 24J as it is incurred. Section 23M limits the deduction where the lender is in a controlling relationship and is not subject to South African tax on the interest	Exclude from the project calculation where the discount rate already reflects the cost of funding
Replacement capital expenditure	Not applicable	A qualifying replacement asset starts a fresh section 12B allowance. The asset replaced gives rise to a recoupment under section 8(4)(a) or an allowance under section 11(o)	Include the full amount as a cash outflow, with its own allowance schedule from the year of first use
Movement in and recovery of working capital	No tax effect	No tax effect	Include as a cash flow only
Disposal at the end of a finite life	Not applicable	The proceeds are not themselves taxable. The recoupment under section 8(4)(a) is taxed at 27 per cent, and any excess of proceeds over cost is a capital gain taxed at 21.6 per cent	Include as a cash outflow in the final year. Exclude from the terminal value, because it is a once-off item

The capital allowance line is the whole of the tax difference between the two alternatives. It is worth its full value only where there is sufficient taxable income to absorb it in years one to three. Under section 20(1)(a) a balance of assessed loss brought forward may be set off only against the higher of R1 million or 80 per cent of taxable income, so allowances that create an assessed loss are deferred rather than lost.